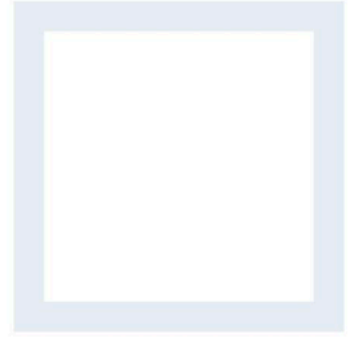


E-commerce Spending Analysis

Aleq Aleqsanyan



The goal is to predict the annual spending of a consumer.

Problem

Predict the total annual spend of each customer

Solution

Study customer data and based on that data, predict their annual spending

Customer Database

```
      Avg. Session Length Time onApp Time on Website
mean          33.053194      12.052488      37.060445
std           0.992563       0.994216       1.010489
min          29.532429       8.508152      33.913847
25%          32.341822      11.388153      36.349257
50%          33.082008      11.983231      37.069367
75%          33.711985      12.753850      37.716432
max          36.139662      15.126994      40.005182
```

```
      Length of Membership Yearly Amount Spent
mean          3.533462      499.314038
std           0.999278       79.314782
min           0.269901      256.670582
25%           2.930450      445.038277
50%           3.533975      498.887875
75%           4.126502      549.313828
max           6.922689      765.518462
```

Data of 2 random customers

Email - hduke@hotmail.com

Avatar - DarkGreen

Adress - 4547 ArcherCommon\nDiazchester

Avg. Session length - 31.926

Time on App - 11.109

Time on Website - 37.268

Length of membership - 2.645

Yearly amount spent - 392.204

Email - pallen@yahoo.com

Avatar - Bisque

Adress - 24645 Valerie Unions Suite 582 nCobbborough

Avg. Session length - 33

Time on App - 11.330

Time on Website - 37.110

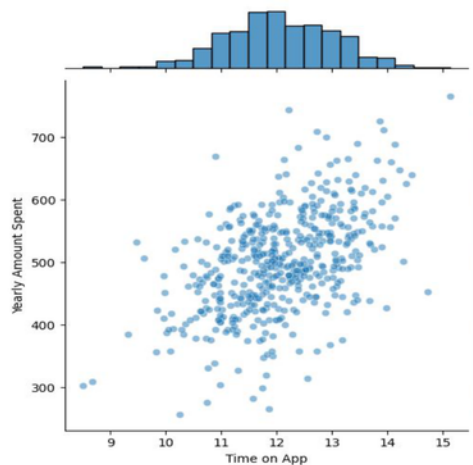
Length of membership - 4.104

Yearly amount spent - 599.406

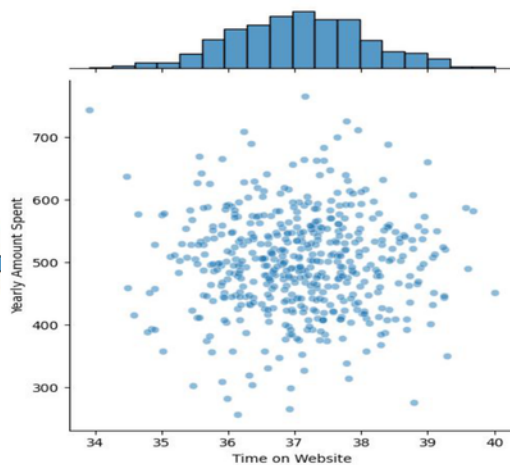
Research Data Analysis: Identifying Key Issues

EDA visualization relationship mapping

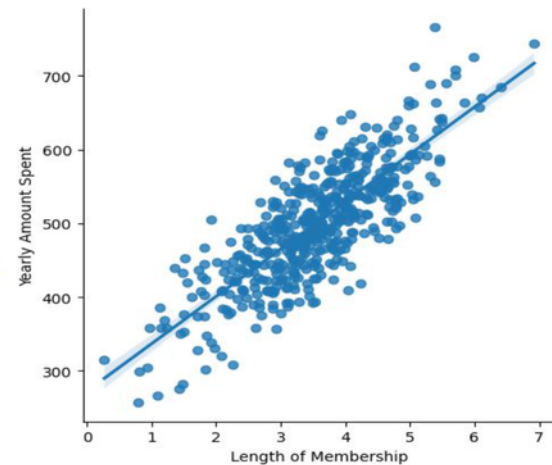
Time on App vs spending



Time on Website vs spending



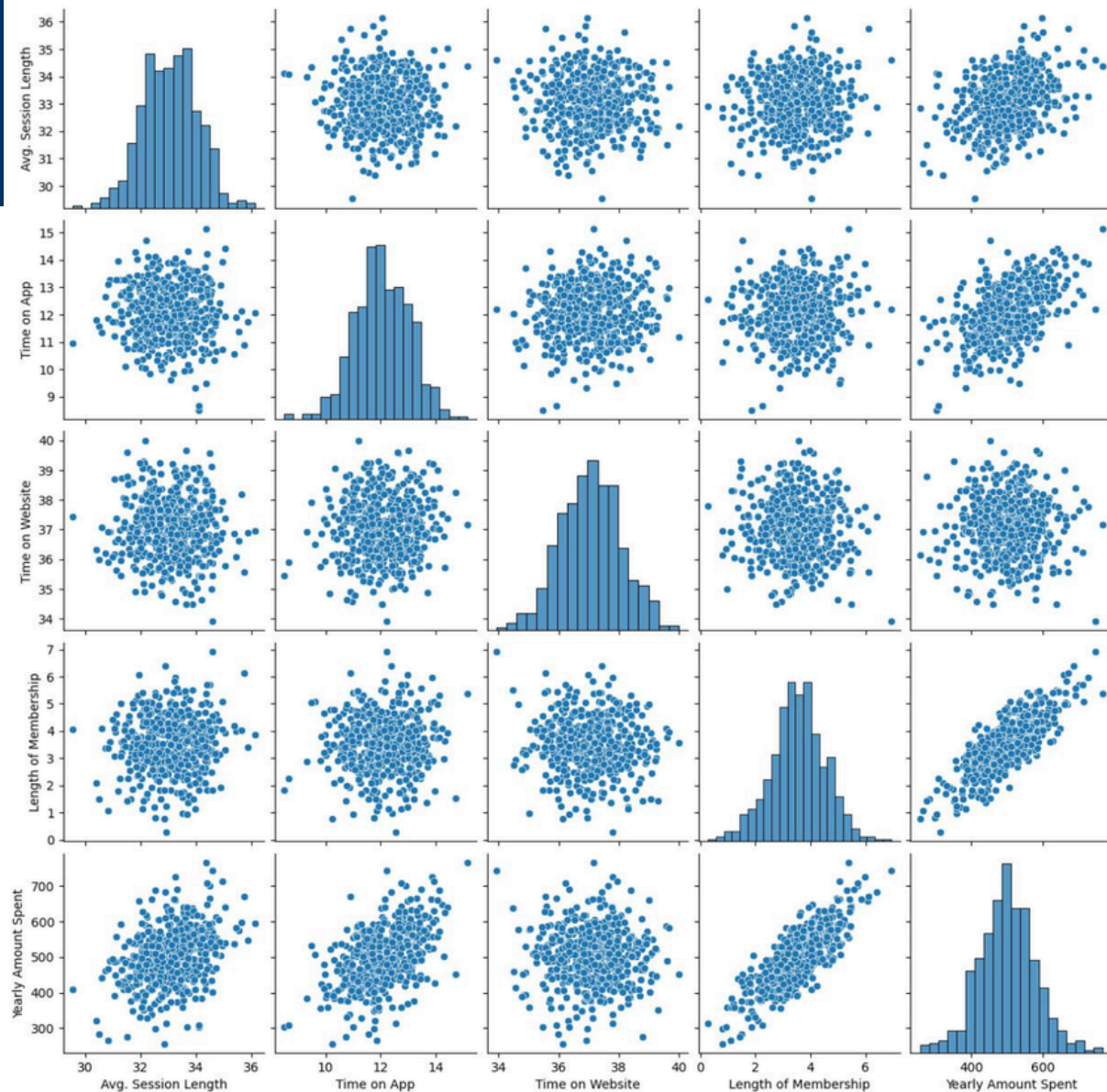
Membership length vs spending



Data investigation

The charts confirm that the highest-spending customers are those with the longest membership history.

Time spent in the app and membership duration show clear growth trends.



Predictive model: Linear Regression

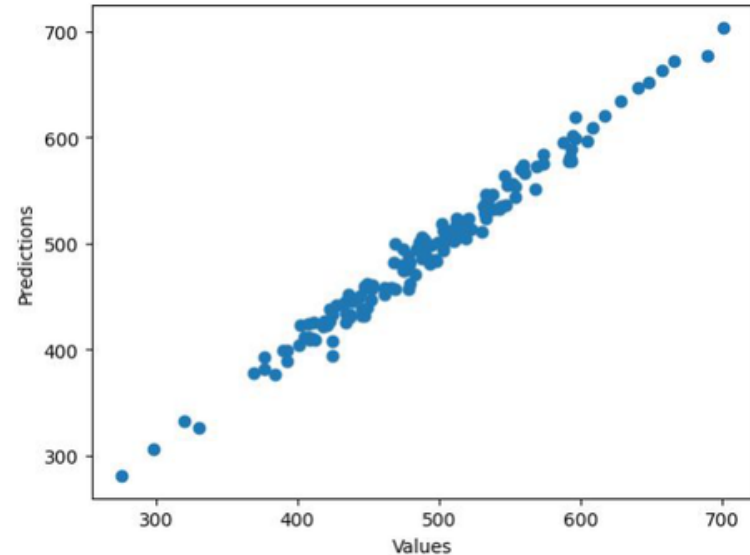
Supervised Learning Approach

I used a Linear Regression model to quantify the specific impact of each customer interaction on the total annual spend. This approach allows us to isolate variables and predict revenue outcomes with statistical significance.

High-precision calculation

The model effectively captures more than 90% of the cost variance.

Values vs Predictions



Model performance: Exceptional accuracy

\$8.42

MAE

Mean Abslue Error
Cost projections are within \$8.42 of the
actual cost.

\$10.19

RMSE

Root Mean Squarred Error
The low population variance confirms the
robust reliability of the model.

✓ The model has high accuracy, which creates a solid foundation for strategic investments.